

HW #4 Software Testing

Com S/SE 4170 Spring 2026

Handed out April 7th, 2026

Questions 1-3 Due at midnight, **Tuesday April 21st as a pdf** uploaded to Canvas

Question 4 is an online survey and is due April 16th

Homework Policy

Homework Policy: The homework assignment should be done individually. You may talk to classmates about the problems in general, but you must complete the homework on your own. You are not permitted to use published answers from websites, etc. Assistance by others must be specifically credited in the solution to the problem that is turned in, describing what the contribution was (e.g., "Thanks to [name] for explaining X in Problem Y", not "Thanks to [name] for help with HW1."). The Dean of Students Office offers several good [resources](#) to build understanding of how to avoid plagiarism, such as Purdue's "[Safe Practices](#)" site.

The goal is for each of you to learn the material sufficiently well to use it productively, to think innovatively, and to develop confidence in your problem-solving abilities. Feel free to talk to me individually about this if you have any questions.

Late policy: 10% penalty per day for late homework. Assignments will not be accepted after April 23rd, unless otherwise arranged/discussed with me. **Note – question 4 is due on April 17th.**

Late Policy One Slack Period for all Homework

During the semester you have one "slack period". This means that you can hand in one homework late – up until the late acceptance date –(e.g. April 23rd for this assignment) without penalty. No questions asked. Once you use up your slack period late penalties will apply as per the late policy

Homework problems are adapted from the course textbook, "Introduction to Software Testing", 2nd edition, Ammann & Offutt, 2017. There is a Student Solution Manuals available online with answers to other practice questions <https://cs.gmu.edu/~offutt/softwaretest/exer-student.pdf>.

Any use of Large Language Models for this homework must be declared.

Q1. Answer the questions based on the following Graph:

$N = \{1, 2, 3, 4, 5, 6\}$

$N_0 = \{1\}$

$N_f = \{5\}$

$E = \{ \{1\ 2\}, \{2\ 3\}, \{2\ 5\}, \{3\ 2\}, \{1\ 4\}, \{4\ 5\}, \{4\ 4\}, \{1\ 6\}, \{6\ 2\}, \{6\ 5\} \}$

And the following candidate test paths:

[1,2,5], [1,6,2,5], [1,4,5]

- (a) **Draw the graph** (you can use the tool from the book website) and include a screenshot/picture (10)
- (b) **How many simple paths are there** (leave out length 0- single nodes)? (5)
- (c) List just the length one paths. **Include the exclamation points and star** on paths you would add if you were creating this by hand.
- (d) **List the prime paths** (10)
- (e) **Does the given set of test paths satisfy** (1) Edge Coverage (2) Edge Pair Coverage (3) Node coverage? If not add test paths to complete those requirements (do not delete existing test paths) (10)

Q2. Draw a control flow graph for the following program fragment. Use the rules from our book (i.e. include extra nodes for loops) (15)

```
c=a;
if (c >0){
    a=a-1
}
if ( a< 1){
    a=a+1
}
else{
    a=a*2
}
while (a > 0) {
    print "inside loop"
    a - -
    if(a ==2) {
        c=c+1;
    }
    c=c+a;
}
print c;
```

Q3. Using a mutation testing tool:

1. Download the tar file from canvas (mutation-pitest-example.tar) and extract on pyrite (this may work on your local machine, but it has been tested on pyrite). This is an example program that uses pitest
2. Move to the triangle-example directory and type
➤ mvn verify

(you may see some warnings but you can ignore those)

3. Once the program has run **copy the output report (on the screen)** starting with:

```
=====
- Timings
=====
(10)
```

Highlight or provide a summary at the end that states the overall statistics (number of total mutants generated and killed) **(5)**

4. Go to the target/pit-reports directory and copy the report directory to your local machine. Open up the index.html file. And take a screenshot (handin) **(10)**
5. Click on com.example and then on triangle.java to show the source code and mutations. Take a screenshot of the mutations for lines 15-25 **(10)**
6. If you hover over the mutants, it will tell you what they are and if they survived or were killed. **Pick ONE mutation that survived** and explain why. Make sure to state **which line/mutation** it is. **(10)**

Q4. List your team members and topic (5 points)

Take the online survey and complete by April 17th